

Test worksheet

Lesson 1

1. The functions f and fg are defined by $f(x) = e^{(x^2 + 3)}$ for $x < 0$ and $fg(x) = e^{(2x)}$ for $x > 3/2$. Explain why f^{-1} exists. [1 mark]

2. Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1} . [5 marks]

3. Hence find and simplify an expression for $g(x)$. [2 marks]

Answer key

1. f is one-to-one (or one-one) on its domain
2. $f^{-1}(x) = -\sqrt{\ln(x) - 3}$, domain $x > e^3$, range $f^{-1} < 0$
3. $g(x) = -\sqrt{2x - 3}$